

A Novel Knowledge Enhanced Large Language Model Augmented Framework for Medical Question Answering

1st Haochen Zou

School of Computer Science and Engineering
Nanjing University of Science and Technology
Nanjing, China
haochenzou@njjust.edu.cn

2nd Yongli Wang

School of Computer Science and Engineering
Nanjing University of Science and Technology
Nanjing, China
yongliwang@njjust.edu.cn

Abstract—Leveraging domain-specific knowledge from pre-trained large language models and knowledge graphs for reasoning in the medical question answering task has emerged as a prominent research field. However, the accuracy of the inference results is restricted by multiple factors, including the quality of analyzed topic entities, the selected inference path in the knowledge graph, and the absence of mutual updating for embedding representations from large language models and knowledge graphs. In this paper, we propose a novel medical question answering framework based on the domain-specific large language model, aiming to enhance the quality of topic entities by implementing the retrieval augmentation technique. Inspired by the concept of chain-of-thought reasoning, we introduce a joint reasoning approach based on analyzed topic entities to facilitate the generation of accurate inference paths. Additionally, we design a unified embedding mechanism that combines representations from both the large language model and graph neural networks, incorporating a pooling operation for predicting answers to input questions. To the best of our knowledge, this work signifies the pioneering efforts in implementing the retrieval augmentation technique and the joint reasoning approach within the context of the medical question answering task. Experimental results on public benchmark datasets demonstrate that the introduced method outperforms state-of-the-art baseline approaches.

Index Terms—Medical question answering, knowledge graph, large language model, graph neural networks

I. INTRODUCTION

Pre-trained large language models have demonstrated remarkable performance in natural language processing tasks, such as question answering, which is a fundamental component for constructing medical conversation systems [1]. While large language models benefit from broad coverage of prior knowledge acquired from extensive training corpora, they exhibit limited capability in domain-specific structured reasoning. The deficiency arises from the heterogeneous diversity and loose organizational architecture that characterize the text data in the training corpora [2]. Knowledge graphs, serving as structured carriers with explicitly represented knowledge, exhibit powerful organizational capabilities in structured reasoning, enabling explainable predictions by providing clear reasoning paths. Numerous studies have

concentrated on leveraging the comprehensive knowledge of pre-trained large language models and knowledge graphs for reasoning in the medical field. [3]. Graph neural networks, as deep learning-based techniques designed to process data with topological structures, can effectively capture the relationships between entities during the modelling process through graph-structured data. Researchers have extensively utilized graph neural networks in combination with knowledge graphs to enhance domain-specific structured reasoning capabilities.

Combining pre-trained large language models and knowledge graphs for inference presents considerable challenges. Firstly, previous studies have been developed in idealized environments where entities mentioned in the question can be directly retrieved from the knowledge graph utilizing identical terms by default [4]. However, in practical health and medical application scenarios, user-generated questions contain non-standard entity representations. The direct retrieval technique is prone to errors in topic entity identification, introducing noise at the initial stage of the analysis process. Secondly, existing approaches make inferences based on the primary topic entity of the question and its immediate neighbours within a few hops [5]. The aforementioned procedure can introduce numerous entity nodes that are semantically irrelevant to the question, especially as the number of topic entities or hops increases. Although several techniques have incorporated the context of the question into the reasoning process, they are primarily designed for multiple-choice question tasks with provided options [6], which are not suitable for the open-ended nature of medical question answering tasks. Furthermore, recent works that combine knowledge from both pre-trained large language models and knowledge graphs for inference tend to apply them separately [3], [7]. The disjointed application fails to facilitate mutual updating or unification of the corresponding feature representations, thereby limiting the effectiveness of structured reasoning. Therefore, an integrated approach is required to enhance the synergy between pre-trained large language models and graph neural networks.

We propose a concise medical question answering framework that incorporates knowledge from both the domain-

specific large language model and the knowledge graph to analyze input questions. The designed framework comprises multiple key enhancements. Firstly, it utilizes a domain-specific pre-trained large language model, fine-tuned using the entity word masking pre-training objective, to generate contextual embeddings for each term in input questions and identify topic entities with the assistance of the conditional random field. Secondly, the architecture leverages the analyzed topic entities to generate a subgraph from the knowledge graph utilizing the retrieval augmentation technique. The developed subgraph is pruned based on relevance scores, employing the head-pooling method to focus on the most pertinent information. Thirdly, the pruned subgraph is embedded through graph neural networks for inference, emphasizing key entities within the subgraph, which is inspired by the concept of chain-of-thought reasoning. Finally, the framework predicts answers by combining feature embeddings from both the large language model and graph neural networks, employing the pooling operation to integrate the aforementioned features. The predicted answer, in the form of triples, is subsequently transferred into the large language model for generating accurate responses. To validate the proposed medical question answering framework, we constructed a health and medical domain knowledge graph utilizing structured data sourced from the vertical medical website. The streamlined reasoning process for input questions is displayed in Fig.1 below. Compared to ChatGPT, an advanced general-domain conversation system, the proposed method generates brief yet accurate responses, as demonstrated in Fig.2.

Our contributions are illustrated in the following aspects:

- 1) A domain-specific pre-trained large language model is introduced and fine-tuned utilizing the entity word masking pre-training objective. It ensures the accurate extraction of topic entities from input questions.
- 2) The retrieval augmentation technique is implemented to analyze topic entities and generate the question-relevant subgraph from the knowledge graph. Subsequently, the subgraph is pruned based on computed relevance scores.
- 3) Graph neural networks are applied to embed the pruned subgraph, with attention to multiple key entities. It draws inspiration from the concept of chain-of-thought reasoning, enhancing the efficacy of the inference procedure.
- 4) Answers are predicted by combining feature embeddings from both the domain-specific large language model and graph neural networks. A pooling operation is utilized to integrate feature embeddings, ensuring a comprehensive and accurate answer prediction.
- 5) Extensive experiments are conducted, comparing the proposed architecture with state-of-the-art baseline methods on public medical benchmark datasets. The experimental results demonstrate the effectiveness of the designed framework with higher evaluation metrics.

II. RELATED WORK

With the advancement of computational power and the introduction of the Transformer architecture, numerous pre-trained large language models, such as Bidirectional Encoder

Representations from Transformers (BERT), Generative Pre-trained Transformers (GPT), and Large Language Model Meta AI (LLaMA) have been developed [8]. These pre-trained large language models undergo supervised training with carefully designed objectives, leveraging extensive training corpora to acquire contextual representations and prior knowledge for downstream natural language processing tasks. Previous works have explored the potential of pre-trained large language models as feasible knowledge bases for reasoning, developing approaches to address question-answering tasks in the general domain [9]. To enhance the explicitness and interpretability of knowledge, multiple studies have integrated structured knowledge from knowledge graphs into pre-trained large language models for inference [3], resulting in satisfactory outcomes.

Yasunaga et al. introduced a joint reasoning approach that leverages knowledge from both large language models and knowledge graphs for question answering [10]. Their application of the inference method is limited to multiple-choice questions, requiring input entities to be selected from the provided options, which may not align with the dynamics of a dialogue scene. Zafar et al. developed a medical question answering system that reasons over large language model embeddings and knowledge graph representations [11]. Similarly, Thambi et al. presented a question answering technique that utilizes weighted graph neural networks for embedding knowledge graphs, benefiting from knowledge obtained through large language models [12]. Existing studies that integrate knowledge from large language models and knowledge graphs are generally constructed under ideal circumstances [13], where entities mentioned in input questions are assumed to be identical to entities in the knowledge graph, facilitating the straightforward generation of subgraphs. However, the above assumption departs from realistic medical question answering conditions. Additionally, existing methods typically center the inference procedure around the topic entity [14], introducing numerous neighbouring nodes that may be semantically irrelevant to the question, leading to suboptimal performance [15].

III. PRELIMINARIES

This section provides a brief overview of the knowledge graph and a fundamental explanation of graph neural networks.

A. Knowledge Graph

A knowledge graph is a multi-relational and directed graph encompassing entities, relations, and factual statements represented as triples [16]. The structure of a triple can be encapsulated as $\text{triple} = \{\text{entity}_1, \text{relation}, \text{entity}_2\}$. Represented as $\mathcal{KG} = (\mathcal{E}, \mathcal{R})$, a knowledge graph $\mathcal{KG}$ involves a set of entity nodes denoted by $\mathcal{E}$ and a set of relation edges defined as $\mathcal{R}$. Each triple $t_1 = \{e_1, r_1, e_2\} \in \mathcal{KG}$ signifies an elemental fact, where $e_1, e_2 \in \mathcal{E}$, and $r_1 \in \mathcal{R}$. General data is characterized by heterogeneous diversity and lacks a strict organizational structure. In contrast, information in knowledge graphs is presented in a well-organized and structured format [17]. The structured nature makes knowledge graphs particularly amenable to processing by deep neural networks, and

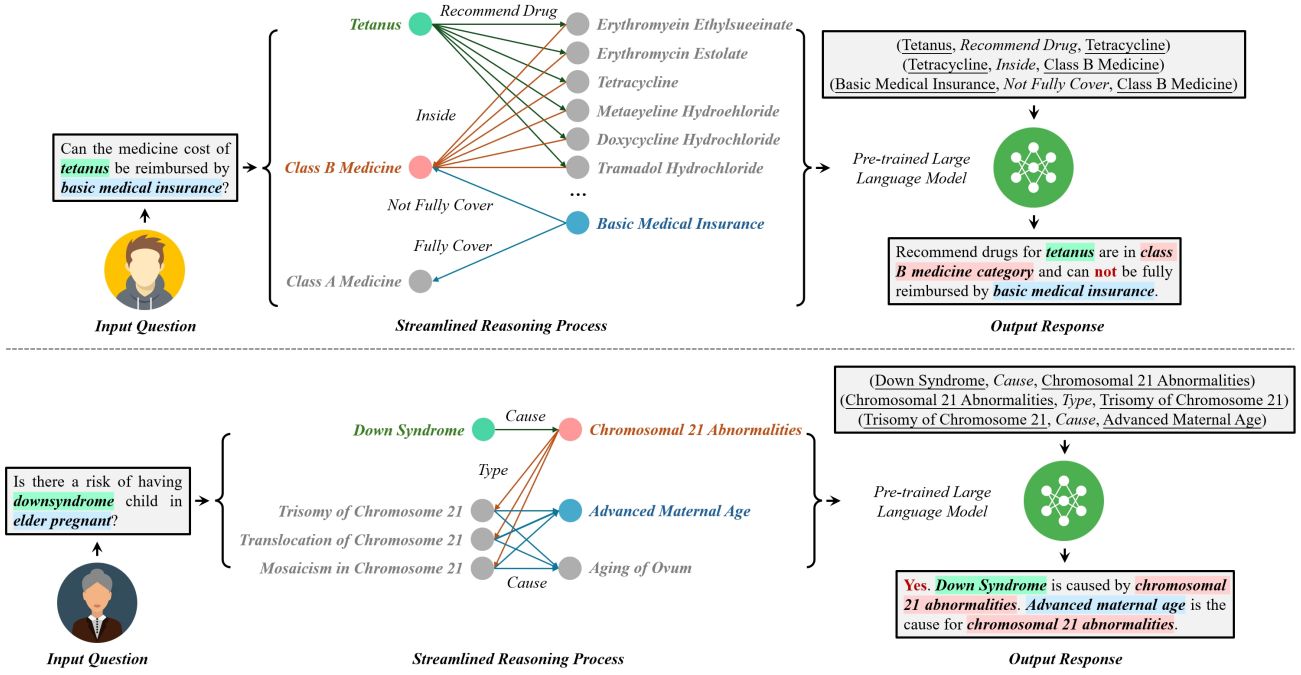


Fig. 1. The streamlined reasoning process for input questions.

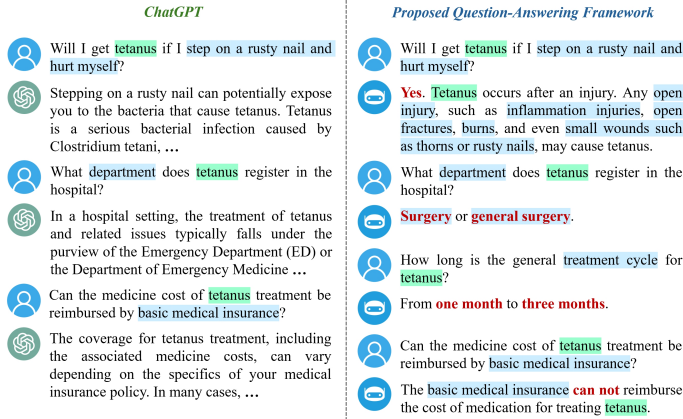


Fig. 2. The comparison of the proposed framework with ChatGPT.

well-suited for tasks comprising structured reasoning, thereby facilitating the generation of explainable predictions.

B. Graph Neural Networks

Structured information in a knowledge graph is encapsulated in the form of entities and relations, defined as triples, to represent the intricate interconnections among entities. To effectively analyze information within a knowledge graph, it is necessary to extract beneficial features from the knowledge graph [13]. Capturing node-level features of entities within the knowledge graph, while preserving the distinct structural characteristics, requires the unveiling of embeddings for the different nodes associated with entities. Embeddings map node-level data into a high-dimensional space, where the distances between points in the space reflect the relationships and

positions of nodes in the original knowledge graph [18]. Graph neural networks offer an effective approach for modelling data with a graph structure and have demonstrated promising results across natural language processing tasks. The fundamental equation for constructing graph neural networks is presented:

$$\mathcal{H}_v^l = \sigma(\mathcal{W}^l \sum_{u \in \mathcal{N}(v)} \frac{\mathcal{H}_u^{l-1}}{|\mathcal{N}(u)|} + \mathcal{B}^l \mathcal{H}_v^{l-1}), \forall l \in \{0, 1, \dots, \mathcal{L}-1\} \quad (1)$$

In the equation, $\mathcal{H}_v^l$ represents the embedding of node v in layer l . The symbol σ signifies a non-linear activation function, such as tanh or ReLU. $\mathcal{W}^l$ is the trainable weight matrix in layer l . The embedding $\mathcal{H}_v^l$ is updated by aggregating the embeddings $\mathcal{H}_u^{l-1}$ from the local neighbouring nodes of node v , denoted as $u \in \mathcal{N}(v)$, from the previous layer $l-1$ utilizing the non-linear activation function σ . $\mathcal{H}_u^{l-1}$ stands for the embedding of node u in the previous layer $l-1$. $\sum_{u \in \mathcal{N}(v)} \frac{\mathcal{H}_u^{l-1}}{|\mathcal{N}(u)|}$ refers to the average of the previous layer embeddings for the neighbouring nodes. $\mathcal{B}^l$ is the trainable bias matrix in layer l . The initial feature of node v , represented as x_v , is considered as the layer 0 embedding $\mathcal{H}_v^0$ of node v .

After running $\mathcal{L}$ iterations of graph neural networks, the embedding for each node v , denoted as $\mathcal{Z}_v$, is obtained below:

$$\mathcal{Z}_v = \mathcal{H}_v^{\mathcal{L}} \quad (2)$$

IV. METHODOLOGY

As illustrated in Fig. 3, the input question is first analyzed utilizing the pre-trained large language model to obtain contextual representations of terms. Entities within the input question

are extracted based on the trained prior knowledge embedded in the large language model. A subgraph of the knowledge graph is gradually retrieved and constructed based on the identified entities using the retrieval augmentation technique. Subsequently, the relevance score for each triple within the subgraph is computed and ranked according to its alignment with the input question. Graph neural networks, built employing the graph attention framework, play an essential role in generating knowledge representations for the subgraph. The answer to the question, in the form of triples, is predicted by combining the contextual representations from the large language model with the knowledge representations derived from graph neural networks. Finally, the response text content of the answer is produced based on the predicted statement triples utilizing the text generation large language model with the assistance of the prompt augmentation method.

A. Entity Identification

The entities in the input question are initially identified as key components for retrieval from the knowledge graph $\mathcal{KG}$. Extracting entities from the text content is defined as a sequential labelling problem, where each term in the text content is labelled as the beginning word **B**, inside word **I**, or outside word **O** of the entity. Pre-trained large language models have demonstrated remarkable performance in fundamental natural language processing tasks in the general domain. However, in specialized fields such as health and medicine, the corresponding performance may decline due to the domain-specific nature of the text content.

The large language model implemented in this research undergoes entity word masking as part of its self-supervised pre-training objectives, deviating from the traditional random word masking mechanism. The aforementioned approach explicitly injects domain-specific knowledge into the language model. The entity words in the training text are masked strategically based on the retrieval augmentation technique, as explained in the following section. The encoding phase is pivotal for capturing contextual representations from the input question, aiding in accurate topic entity identification. In this study, LLaMA serves as the backbone large language model, which is fine-tuned utilizing the entity word masking pre-training objective to generate contextual embeddings for each term in the input question. Each embedding is a high-dimensional vector that represents a term in relation to its surrounding terms, capturing semantic relationships and contextual information. It enables the framework to comprehend the meaning and nuances associated with each term in the input question.

The generated contextual embeddings are utilized as input for the conditional random field during the label decoding process. In this research, the conditional random field is implemented for sequentially labelling each term to generate an accurate sequence of labels. The objective of the conditional random field is to identify the most probable sequence of labels l given the sequence of contextual embeddings $\mathcal{E}^{cont}$. The probability can be computed using the following equation:

$$\hat{l} = \arg \max_l P(l|\mathcal{E}^{cont}) \quad (3)$$

Where the conditional probability $P(l|\mathcal{E}^{cont})$ in a linear-chain conditional random field is calculated in the equation:

$$P(l|\mathcal{E}^{cont}) = \frac{1}{Z(\mathcal{E}^{cont})} \exp\left(\sum_{i=1}^n \theta_f f_i(\mathcal{E}^{cont}, y)\right) \quad (4)$$

In the equation, $Z(\mathcal{E}^{cont})$ represents the normalization factor. n signifies the total number of tokens in the input question. θ_f denotes the learnable feature weights. $f_i(\cdot)$ stands for the feature function, which captures the relationship between the input contextual embeddings $\mathcal{E}^{cont}$ and the output labels l . It transitions relationships between adjacent labels in the sequence. The feature function $f_i(\cdot)$ and the feature weights θ_f are learned from the training corpus by minimizing the negative log-likelihood of the true label sequences.

B. Subgraph Generation

After identifying entities from the question content, it is essential to highlight analyzed entities within the knowledge graph $\mathcal{KG}$ to generate a question-specific subgraph $\mathcal{KG}_{sub}$. Considering the entire knowledge graph $\mathcal{KG}$ with all entities and corresponding relations for further analysis can be both time-consuming and unnecessary, given the massive size of the knowledge base. The creation of the subgraph $\mathcal{KG}_{sub}$ from the knowledge graph $\mathcal{KG}$ serves to reduce computation time by constraining the number of candidates to be searched.

Considering practical situations, the descriptions of the identified entities from the input question may not be entirely identical to the entities in the knowledge graph, even if they convey the same meaning. When the description of an entity is vague, utilizing direct matching, as in previous studies, can introduce additional noise information, thereby reducing the accuracy of subsequent model operations. To address the aforementioned issue, this research incorporates the retrieval augmentation technique. It involves encoding and storing the text data of entities and relations from the knowledge graph in the vector database. Given a candidate entity e_i from the input question, the retrieval augmentation technique first transforms the candidate entity e_i into its embedding vector. Subsequently, it compares the embedding vector and extracts the potential standardized entity e'_i in the knowledge graph $\mathcal{KG}$ from the vector database based on the highest semantic similarity score.

The research employs the Large Language Model Embedder (LLM-Embedder)¹, a unified embedding mechanism, to convert knowledge information into embedding vectors and compare semantic similarity scores with the candidate entity e_i from the input question. Cosine similarity is utilized to measure the semantic similarity between the embedding vector of the candidate entity e_i analyzed from the input question and the embedding vector of the potential standardized entity e'_i from the knowledge graph. Additionally, Manhattan distance and Euclidean distance are used to examine semantic similarity

¹<https://huggingface.co/BAAl/llm-embedder/>

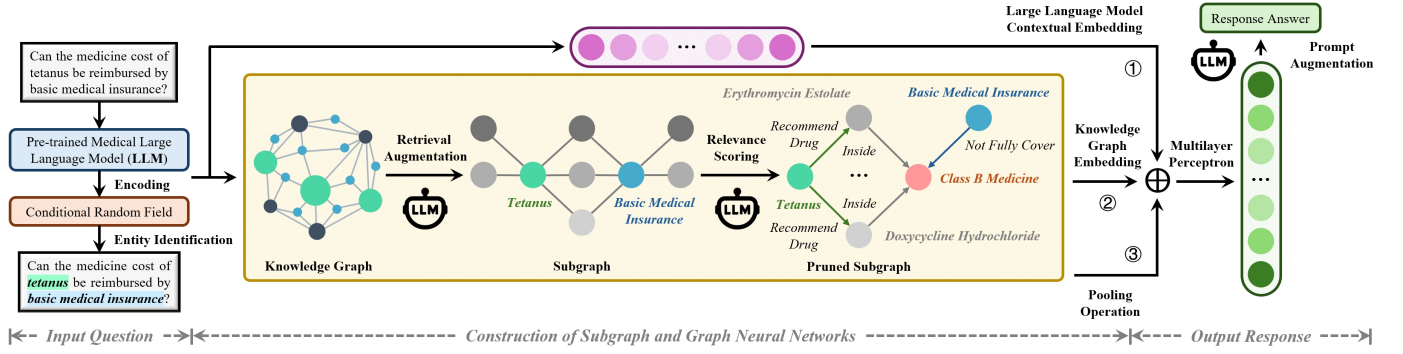


Fig. 3. The overview of the knowledge-enhanced large language model augmented medical question answering model.

between the two embedding vectors. Experimental results from existing studies indicate that these measurement approaches generate approximately similar results [19]. By leveraging the pre-trained large language model, the LLM-Embedder model combines both generality and specialty based on large-scale training datasets with robust training workflows, showcasing remarkable performance across various natural language processing tasks, particularly in the retrieval augmentation technique. The construction of the subgraph $\mathcal{KG}_{sub}$ involves the entities and relations mentioned in the input question, which are refined using the retrieval augmentation technique.

C. Subgraph Pruning

Numerous entity nodes within the subgraph $\mathcal{KG}_{sub}$ that are heuristically retrieved from the knowledge graph $\mathcal{KG}$ can be irrelevant to the context of the question-and-answer conversation. The inclusion of few-hop neighbour nodes of the retrieved entity nodes in the subgraph $\mathcal{KG}_{sub}$ can introduce entities that are uninformative for the inference procedure, which creates unnecessary complexity in subsequent reasoning tasks, especially when confronted with a substantial collection of retrieved entities, potentially leading to overfitting and hindering the smooth progression of inference. To streamline the process and enhance the efficiency of further analyses, it is essential to eliminate and filter out irrelevant nodes. To address the aforementioned concern, a node relevance ranking mechanism is proposed. It employs a domain-specific pre-trained large language model to score the relevance of each entity node in the knowledge graph $\mathcal{KG}$, conditioned on the context of the question-and-answer conversation.

For each node e_j in the set of entity nodes $\mathcal{E}_{sub}$ extracted from the subgraph $\mathcal{KG}_{sub}$, the textual description t_e associated with the entity node e_j is concatenated with the context of the input question t_Q . The relevance score ρ_{e_j} for pruning the subgraph $\mathcal{KG}_{sub}$ is computed using the following equation:

$$\rho_{e_j} = \text{Pool}(f_{enc}(\mathcal{W}_t(t_{[CLS]}; t_Q; t_{[SEP]}; t_{e_j}; t_{[SEP]}) + \mathcal{B}_t)) \quad (5)$$

In the equation for pruning the subgraph $\mathcal{KG}_{sub}$, the head-pooling operation $\text{Pool}(\cdot)$ is employed to aggregate the outcome of the [CLS] token, which serves as a contextualized

representation for the sequence derived from the concatenated text content. f_{enc} represents the encoder function, typically a part of the large language model, which processes the input sequence. $\mathcal{W}_t$ is the learnable weight matrix. $t_{[CLS]}$ signifies a special token added as the beginning token of each comparison text content. The hidden state of $t_{[CLS]}$ at the output of the encoder denotes the representation of the entire question-and-answer conversation text. $t_{[SEP]}$ refers to a special token introduced to mark the end of the text content, delineating different parts of the input sequence. $\mathcal{B}_t$ stands for the learnable bias matrix.

The relevance score ρ_{e_j} signifies the significance of each entity node e_j within the subgraph $\mathcal{KG}_{sub}$ relative to the contextual information provided by the question-and-answer conversation. It plays a crucial role in the reasoning process and in pruning the subgraph $\mathcal{KG}_{sub}$, aiding in identifying and retaining key entities relevant to the input question.

D. Subgraph Embedding

The process of subgraph embedding relies on graph neural networks, which utilize deep learning techniques to process data with topological structures. Graph neural networks demonstrate satisfactory performance and traceability, making them widely applicable across various research fields. They effectively capture the relationships between entities in the modelling process by representing graph-structured data. Graph neural networks designed in this research are built upon the graph attention framework. In the architecture, each entity node undergoes a process where it receives messages from its nearby incoming neighbour nodes. Additionally, an attention mechanism is introduced in conjunction with the initialized entity relation embeddings. Specifically, the relevance scores ρ_{e_j} obtained for entity nodes are incorporated by concatenating relevance scores with embedding features of entity nodes.

The embedding $\mathcal{H}_{e_j}^{l+1}$ of an entity node $e_j \in \mathcal{E}_{sub}$ with the relation $r \in \mathcal{R}_{e_j}$ from the subgraph $\mathcal{KG}_{sub}$ at layer $l+1$ is shown in the equation below:

$$\mathcal{H}_{e_j}^{l+1} = \sigma\left(\sum_{r \in \mathcal{R}_{e_j}} \sum_{e_k \in \mathcal{N}_{e_j}^r} \alpha_{\{e_j, e_k\}} \frac{1}{|\mathcal{N}_{e_j}^r|} \mathcal{W}_r^l \mathcal{H}_{e_k}^l + \mathcal{W}_0^l \mathcal{H}_{e_j}^l\right) \quad (6)$$

Where $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a 2-layer multilayer perceptron (MLP) with batch normalization. $\mathcal{R}_{e_j}$ is the set of relations associated with the entity node e_j . $\mathcal{N}_{e_j}^r$ stands for the set of neighbour nodes of entity node e_j that are connected through the relation $r \in \mathcal{R}_{e_j}$. $\alpha_{\{e_j, e_k\}}$ represents the normalized attention weight for entity node e_j and entity node e_k . $\mathcal{W}_r^l \in \mathbb{R}^{d+1}$ refers to the trainable weight matrix at the relation level. $\mathcal{H}_{e_k}^l \in \mathbb{R}^d$ denotes the embedding of the previous layer for neighbour node e_k . For every entity node $e_j \in \mathcal{E}_{sub}$, the initial representation $\mathcal{H}_{e_j}^0$ at layer 0 is determined by applying a linear transformation that maps the initial node embedding of the entity node e_j to a vector space $\mathbb{R}^d$, where d represents the desired dimensionality of the embedding.

The attention mechanism plays an essential role in capturing the strength of association between two entity nodes, informed by the combination of factors such as node types, relations, and node relevance scores. It helps weigh the importance of different relationships and entities, contributing to a nuanced and context-aware understanding of the knowledge graph structure. In this research, a single-layer feedforward neural network is implemented to compute the attention weights. The calculation involves a linear transformation weight matrix $\mathcal{W}$, and the relevance score ρ_{e_j} , which is shared across all entity nodes within the subgraph $\mathcal{KG}_{sub}$. The nonlinearity chosen is LeakyReLU with a negative input slope of $\alpha = 0.2$. The attention weight $a_{\{e_j, e_k\}}$ from entity node e_j to entity node e_k is calculated using the following equation:

$$a_{\{e_j, e_k\}} = \text{LeakyReLU}(\rho_{e_j}[\mathcal{W}_{e_j}, \mathcal{W}_{e_k}]) \quad (7)$$

For a given entity node e_j , the computation of attention weights is restricted to its directly connected neighbour nodes. Attention weights are subsequently normalized utilizing the SoftMax activation function, resulting in the equation below:

$$\alpha_{\{e_j, e_k\}} = \frac{\exp(a_{\{e_j, e_k\}})}{\sum_{e_m \in \mathcal{N}_{e_j}^r} \exp(a_{\{e_j, e_m\}})} \quad (8)$$

E. Answer Prediction

Given an input question Q and a set $\mathcal{T}$ containing multiple answer triples $t = (e_1, r_1, e_2)$, the goal of the proposed method is to leverage contextual and feature representations from both the large language model and the knowledge graph to calculate the probability of a specific set of triples being the correct answer, as displayed in the following equation:

$$P(Q|\mathcal{T}) \propto \exp(\text{MLP}(\mathcal{E}^{\text{LLM}}, \mathcal{H}^{\text{GNN}}, \text{Pool}(\mathcal{H}_{e_j}^l))) \quad (9)$$

In the equation, $\mathcal{E}^{\text{LLM}}$ denotes the embedding of the input question Q produced by the domain-specific pre-trained large language model, which leverages its prior knowledge. $\mathcal{H}^{\text{GNN}}$ represents the embedding of all entity nodes $e_1, e_2, \dots, e_n \in \mathcal{E}_{sub}$ within the subgraph $\mathcal{KG}_{sub}$, which are obtained after the entity nodes undergo encoding and pass through $\mathcal{L}$ layers of graph neural networks. $\text{Pool}(\mathcal{H}_{e_j}^l)$ stands for the pooling operation applied to the set of embeddings $\{\mathcal{H}_{e_j}^l | e_j \in \mathcal{E}_{sub}\}$.

TABLE I
THE INFORMATION OF FIVE STANDARD BENCHMARK DATASETS.

Datasets	Training	Validation	Testing	Evaluation Metric
MEE	15,000	5,000	3,000	Micro-F1
MIE	14,339	3,585	4,482	Micro-F1
QIC	6,931	1,955	1,994	Accuracy
STS	16,000	4,000	10,000	Micro-F1
QQR	15,000	1,600	1,596	Accuracy

During the training process, each input question Q is associated with a set of candidate entities, with a limitation of 5 candidates. The optimization process is end-to-end, meaning that both the large language model and the components of graph neural networks are trained jointly. The cross-entropy loss is utilized as the optimization objective, facilitating the framework to learn and improve its performance by minimizing the difference between predicted and actual outcomes.

V. EXPERIMENTS

A. Datasets and Experimental Settings

To evaluate the efficacy of the framework in the medical question answering task, we first construct a knowledge graph utilizing data obtained from the doctor-patient communication platform, XunYiWenYao, which serves as a professional health and medicine encyclopedia [20]. The mined data, formatted as JSON files, encompass 8,808 types of diseases, each categorized into 23 detailed categories. These categories include information such as the disease name, definition, prevention measures, causes, etc. During the development of the knowledge graph, health and medical entities serve as nodes, while the detailed category descriptions act as relation types. The knowledge graph is constructed employing the Neo4j database, a robust tool for managing and querying graph data. The domain-specific pre-trained large language model implemented in this research is the Llama-Medical-Chatbot². It is based on the backbone large language model Llama-2-7B³, and has been fine-tuned on the textbook named The Gale Encyclopedia of Medicine. Consequently, health and medical knowledge are both implicitly encoded in the large language model and explicitly represented in the knowledge graph.

We conduct thorough experiments on five publicly available standard benchmark datasets to assess the medical information processing and reasoning capabilities of the proposed question answering framework. The benchmark datasets encompass a diverse range of medical tasks, including the Medical Entity Extraction (MEE) dataset, the Medical Information Extraction (MIE) dataset, the Query Intent Criterion (QIC) dataset, the Semantic Textual Similarity (STS) dataset, and the Query-Query Relevance (QQR) dataset [21]. The detailed information for the five benchmark datasets is provided in Table I.

We present the experimental and hyperparameter settings of the developed framework. The domain-specific large language model undergoes fine-tuning based on the designed entity

²<https://huggingface.co/ThisIs-Developer/Llama-2-GGML-Medical-Chatbot/>.

³<https://huggingface.co/meta-llama/Llama-2-7b/>.

TABLE II
THE EXPERIMENTAL RESULTS ON BENCHMARK DATASETS ARE
COMPARED WITH BASELINE APPROACHES.

Models	MEE	MIE	QIC	STS	QQR
Conco-ERNIE	0.689	0.657	<u>0.873</u>	<u>0.865</u>	<u>0.863</u>
Medical-ChatGLM	0.673	0.628	0.846	0.853	0.851
HCT-MacBERT	0.649	0.595	0.835	0.808	0.812
CEMR-LM	0.667	0.633	0.857	0.848	0.837
Medical-DKPLM	0.658	0.619	0.842	0.833	0.845
KG-BERT	0.697	0.701	0.849	0.827	0.839
KIMedQA	0.683	0.665	0.830	0.822	0.841
GNN-BERT	0.676	0.637	0.834	0.815	0.835
Proposed Method	0.735	0.727	0.895	0.878	0.892

word masking pre-training objective. The Transformer stack comprises 12 Transformer decoders, each with 12 attention heads. For hyperparameter settings, the embedding dimension is set to 2046. The Adam optimizer is employed, with a learning rate of 5e-5. The training process extends over 50 epochs, utilizing a dropout rate of 0.2 and a batch size of 64. The framework is implemented using the PyTorch platform, version 1.12. The Python version is 3.8, and the Transformer version is 4.35.2. All experiments are conducted on a server with the Ubuntu 23.0 system, 128GB RAM, an Inter Core 13900K CPU processor @5.80 GHz, and a 48GB NVIDIA GeForce RTX A6000 GPU card.

B. Performance Evaluation

We introduce the baseline methods compared in the experiment, including five domain-specific large language model-based approaches and three graph neural network-based techniques. The five large language model-based baseline competitors are Conco-ERNIE [22], Medical-ChatGLM [23], HCT-MacBERT [24], CEMR-LM [25], and Medical-DKPLM [26]. The three graph neural network-based techniques are KG-BERT [27], KIMedQA [11], and GNN-BERT [12].

The experimental results are presented in Table II. In the presentation of results, the best outcomes are highlighted in bold font, and the second-best results are underlined. Upon reviewing the displayed results, it is evident that incorporating structured domain-specific knowledge from the knowledge graph significantly enhances the performance of pre-trained language models in information extraction tasks, such as named entity recognition and entity relation identification. The improvement is attributed to the utilization of structured information retrieved from the knowledge graph. In natural language processing tasks that require comprehending contextual semantics to make judgments, such as query intent criterion, semantic textual similarity, and query-query relevance, the performance of various baseline methods primarily depends on the scale of the parameters within the backbone pre-trained large language models. The domain-specific fine-tuning procedure also proves beneficial, as it helps the backbone large language models build robust prior knowledge, which contributes to tasks focused on contextual semantics.

In comparison to the baseline methods, the proposed framework demonstrates a significant improvement in information

TABLE III
THE EXPERIMENTAL RESULTS OF ABLATION STUDY.

Models	MEE	MIE	QIC	STS	QQR
w/o Pre-training Objective	0.712	0.703	0.870	0.862	0.879
w/o Retrieval Augmentation	0.695	0.698	<u>0.875</u>	<u>0.871</u>	0.880
w/o Relevance Score Pruning	<u>0.720</u>	<u>0.715</u>	0.868	0.859	0.865
w/o Subgraph Embedding	0.699	0.707	0.860	0.855	0.857
Proposed Method	0.735	0.727	0.895	0.878	0.892

extraction tasks, achieving approximately a 3% increase over the second-best competitors. The introduction of the entity word masking pre-training objective during fine-tuning and the implementation of the retrieval augmentation technique contribute to the ability of the architecture to accurately identify entities and relations from the input question based on the constructed knowledge graph. The proposed framework exhibits improved performance in tasks related to inference and comprehending contextual semantics from the input question, showcasing approximately a 2% increase over the second-best competitors. The enhancement is primarily due to the design of the joint reasoning mechanism, which makes predictions by combining embeddings from both the large language model and graph neural networks, incorporating a pooling operation. It comprehensively leverages the strengths of prior knowledge and structured knowledge. Consequently, the architecture proves to be capable of reasoning tasks, such as analyzing the intent behind questions from patients and detecting whether requests from different patients pertain to the same topic.

C. Ablation Study

The experimental results of the ablation study are displayed in Table III. In the presentation of results, the best outcomes are highlighted in bold font, and the second-best results are underlined. Without is abbreviated as w/o in the table.

The results of the ablation study reveal that withdrawing key components of the framework leads to noticeable declines in overall performance across all benchmark datasets. The most significant decrease in information extraction tasks occurs when the retrieval augmentation technique is removed, underscoring its importance in incorporating structured health and medical domain knowledge into the architecture. When the embedding generated by graph neural networks from the subgraph of the knowledge graph is excluded, and only the embedding representations from the pre-trained large language model are utilized, the performance of the framework in tasks related to reasoning based on contextual content experiences a substantial decline. It emphasizes the value of utilizing the reasoning path through the knowledge graph, as subgraph embeddings enhance the inference procedure. The removal of the relevance score pruning approach for information extraction tasks results in a limited decline, suggesting that while it contributes to the inference process, its impact is less effective than that of the retrieval augmentation technique. Conversely, the retrieval augmentation technique plays a more critical role in information extraction tasks compared to tasks involving

inference and comprehending contextual semantics from the input question.

The removal of the retrieval augmentation technique results in a slower rate of convergence during the validation process among all benchmark datasets. The observation underscores the role of structured health and medical domain knowledge in expediting inference across various natural language processing tasks. Despite the incorporation of the entity word masking pre-training objective and relevance score pruning may increase time costs for development and addressing downstream tasks, the improved performance demonstrated in the ablation study results suggests that the key components are valuable additions to the proposed framework.

VI. CONCLUSION

In this research, we introduce a medical question answering framework that significantly enhances inference capabilities. The improvement is achieved through the precise extraction of question topic entities from the knowledge graph, utilizing a domain-specific large language model fine-tuned with the customized fine-tuning procedure, combined with the retrieval augmentation technique. The generated subgraph is further refined by pruning based on computed relevance scores. The subgraph is subsequently fed into graph neural networks to produce embeddings that incorporate inference paths from key entities. The architecture is further enhanced by predicting answers to input questions through a joint reasoning approach, which leverages embeddings from both the large language model and graph neural networks, together with a pooling operation. Experimental results demonstrate that the proposed framework outperforms state-of-the-art competitors, establishing a robust baseline for future studies.

ACKNOWLEDGMENT

This paper is partially supported by the Key Medical Research Projects of Jiangsu Provincial Health Commission (ZD2022068), Jiangsu Province Key R&D Program (Modern Agriculture) Key Project (BE2023352), Key Project of Philosophy and Social Science Research in Universities of Jiangsu Province (2024SJZD085), Jiangsu Sports Strength Province Construction Research Center Project (24SSL122).

REFERENCES

- [1] X. Luo, Z. Deng, and B. Yang, "Pre-trained language models in medicine: A survey," *Artificial Intelligence in Medicine*, p. 102904, 2024.
- [2] M. S. Jahan, and M. Oussalah, "A systematic review of Hate Speech automatic detection using Natural Language Processing," *Neurocomputing*, p. 126232, 2023.
- [3] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [4] B. Shao, X. Li, and G. Bian, "A survey of research hotspots and frontier trends of recommendation systems from the perspective of knowledge graph," *Expert Systems with Applications*, vol. 165, p. 113764, 2021.
- [5] X. Zhang, J. Dang, Y. Wang, and S. Li, "Meta-learning framework with updating information flow for enhancing inductive prediction," *Knowledge-Based Systems*, vol. 294, p. 111720, 2024.
- [6] Q. Zhang, S. Chen, M. Fang, and X. Chen, "Joint reasoning with knowledge subgraphs for Multiple Choice Question Answering," *Information Processing & Management*, vol. 60, no. 3, p. 103297, 2023.
- [7] L. Yang, H. Chen, Z. Li, X. Ding, and X. Wu, "Give us the facts: Enhancing large language models with knowledge graphs for fact-aware language modeling," *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [8] H. Zhang, H. Song, S. Li, M. Zhou, and D. Song, "A survey of controllable text generation using transformer-based pre-trained language models," *ACM Computing Surveys*, vol. 56, no. 3, pp. 1–37, 2023.
- [9] L. Hu, Z. Liu, Z. Zhao, L. Hou, L. Nie, and J. Li, "A survey of knowledge enhanced pre-trained language models," *IEEE Transactions on Knowledge and Data Engineering*, 2023.
- [10] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, and J. Leskovec, "QA-GNN: Reasoning with language models and knowledge graphs for question answering," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 535–546, 2021.
- [11] A. Zafar, S. K. Sahoo, D. Varshney, A. Das, and A. Ekbal, "KIMedQA: towards building knowledge-enhanced medical QA models," *Journal of Intelligent Information Systems*, pp. 1–26, 2024.
- [12] S. V. Thanbi, and P. C. Reghu Ra, "A novel technique using graph neural networks and relevance scoring to improve the performance of knowledge graph-based question answering systems," *Journal of Intelligent Information Systems*, pp. 1–24, 2024.
- [13] C. Peng, F. Xia, M. Naseriparsa, and F. Osborne, "Knowledge graphs: Opportunities and challenges," *Artificial Intelligence Review*, vol. 56, no. 11, pp. 13071–13102, 2023.
- [14] J. Bai, X. Liu, W. Wang, C. Luo, and Y. Song, "Complex query answering on eventuality knowledge graph with implicit logical constraints," *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [15] L. Liu, M. Liu, S. Liu, and K. Ding, "Event extraction as machine reading comprehension with question-context bridging," *Knowledge-Based Systems*, p. 112041, 2024.
- [16] X. Liu, T. Mao, Y. Shi, and Y. Ren, "Overview of Knowledge Reasoning for Knowledge Graph," *Neurocomputing*, p. 127571, 2024.
- [17] L. Zhong, J. Wu, Q. Li, H. Peng, and X. Wu, "A comprehensive survey on automatic knowledge graph construction," *ACM Computing Surveys*, vol. 56, no. 4, pp. 1–62, 2023.
- [18] L. Waikhom, and R. Patgiri, "A survey of graph neural networks in various learning paradigms: methods, applications, and challenges," *Artificial Intelligence Review*, vol. 56, no. 7, pp. 6295–6364, 2023.
- [19] P. Zhang, S. Xiao, Z. Liu, and Z. Dou, "Retrieve anything to augment large language models," *arXiv preprint arXiv:2310.07554*, 2023.
- [20] C. Liu, H. Sun, N. Du, S. Tan, and H. Fei, "Augmented LSTM framework to construct medical self-diagnosis android," in *Proceedings of the 2016 IEEE 16th International Conference on Data Mining*, pp. 251–260, 2023.
- [21] N. Zhang, M. Chen, Z. Bi, X. Liang, and L. Li, "Cblue: A Chinese biomedical language understanding evaluation benchmark," *arXiv preprint arXiv:2106.08087*, 2022.
- [22] B. Zhang, Z. Tu, S. Hang, D. Chu, and X. Xu, "Conco-ERNIE: Complex User Intent Detect Model for Smart Healthcare Cognitive Bot," *ACM Transactions on Internet Technology*, vol. 23, no. 1, pp. 1–24, 2023.
- [23] L. Fan, X. Liu, Y. Wang, G. Yang, Z. Du, and G. Wang, "Enhancing Medical Language Understanding: Adapting LLMs to the Medical Domain through Hybrid Granularity Mask Learning," in *Proceedings of the 2023 IEEE International Conference on Bioinformatics and Biomedicine*, pp. 2990–2995, 2023.
- [24] M. Wang, X. He, L. Liu, Q. Fang, M. Zhang, H. Chen, and Y. Liu, "HCT: Chinese medical machine reading comprehension question-answering via hierarchically collaborative transformer," *IEEE Journal of Biomedical and Health Informatics*, 2024.
- [25] Y. Zou, P. Zhang, Y. Ling, D. Lv, Z. Li, S. Lu, and G. Zhang, "A domain adaptive pre-training language model for sentence classification of Chinese electronic medical record," in *Proceedings of the 2023 IEEE International Conference on Bioinformatics and Biomedicine*, pp. 4776–4783, 2023.
- [26] T. Zhang, C. Wang, N. Hu, M. Qiu, C. Tang, and X. He, "DKPLM: decomposable knowledge-enhanced pre-trained language model for natural language understanding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 10, pp. 11703–11711, 2022.
- [27] Q. Guo, S. Cao, and Z. Yi, "A medical question answering system using large language models and knowledge graphs," *International Journal of Intelligent Systems*, vol. 37, no. 11, pp. 8548–8564, 2022.